

Case B — NV Triangle Closure with 8.5× Thermal Tightening

Showcase 2 of the Eikonal-to-Entanglement Demo Suite

Eikonal Bridge - Showcases used for Colloquium @ NCU Physics, April 28, 2026

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Methodology Note: how to read this showcase

This showcase follows the same seven-slot grammar as Case A, with the binary bridge identity $S = F$ now generalised to a three-leg triangle closure in which one constraint leg (optical), one data leg (detection), and one environmental leg (thermal) are tied together by the NV^- zero-field splitting coefficient dD/dT .

- Slot (i) names the canonical physical system: an NV^- centre in bulk diamond with a GaN solid-immersion lens.
- Slot (ii) states the mathematical structure: three multiplicative fidelity legs; three disambiguated tightening symbols κ_{target} , κ_{budget} , κ_{physical} ; and the H_C identification theorem stating $\text{cond}(H_C)|_{c=0} = \kappa_{\text{target}}^2$ under mixed-budget parameterisation.
- Slot (iii) describes the four verification channels D1–D4.
- Slot (iv) presents the correspondence check at the anchor ($T_2^* = 3\mu\text{s}$, $\text{NA} = 0.95$, $\Delta T_{\text{lab}} = 10\text{mK}$).
- Slot (v) stress-tests the triangle along three sweep axes and maps $\text{cond}(H_C)$ on the $(T_2^*, \Delta T_{\text{lab}})$ plane.
- Slot (vi) states the caveats.
- Slot (vii) states the identification-theorem novelty claim and the Gap 2 falsifying-measurement prediction.

Pedagogical payload. The Slot (vii) claim is a *theorem* in the mixed-budget parameterisation of Slot (ii), not a numerical coincidence. This means the Gap 2 test becomes a measurement of the cross-coupling residual $\text{cond}(H_C)_{\text{meas}}/\kappa_{\text{target,meas}}^2 - 1$, which is a physically interpretable cross-coupling signature rather than an ad-hoc numerical match.

Atlas Anchor Box

Inheritance tags. Case A §Slot(ii) (bridge identity $S = F = |\mathcal{M}[\phi]|^2$); PC4 §Slot(iii) (Richards–Wolf high-NA quadrature for the SIL leg); PC7 §Slot(i) (Hessian-based tolerance allocation); PC9 §Slot(ii)–(iii) (dephasing tightening $\zeta = e^{\sigma_{\phi,d}^2}$ and inverse-FFT PSD Monte Carlo reused as the thermal leg).

Reused artefacts. Companion script `case_B_figures_v1p2.py` is the authoritative source of every numerical value cited in Tables 1 and 3, and in the Conclusion. `LITERATURE_ANCHORS.md` §3 supplies all experimental anchors; `dblgauss_f2p8.seq` supplies the canonical SIL-plus-objective prescription inherited from Case A.

List of Abbreviations

Abbr.	Expansion	Context in this document
CP	Code Page (CP950, CP1252)	Windows encoding compliance
CPTP	Completely Positive Trace-Preserving	Kraus channel framework (PC11, Book 2)
CRB	Cramér–Rao Bound	QFI forward pointer (PC5)
D1–D4	Derivation / verification channels	four-channel correspondence (Slot iv)
FFT	Fast Fourier Transform	inverse-FFT PSD sampling (D3)
GaN	Gallium Nitride	SIL material ($n = 2.40$)
LCB	Literature Consistency / Benchmark	Table 3 row typing
MC	Monte Carlo	D3 ensemble draws (Slot v)
MW	Microwave	2.87 GHz NV spin drive
NA	Numerical Aperture	objective parameter (0.95)
NV	Nitrogen-Vacancy colour centre	NV [−] charge state assumed throughout
ODMR	Optically Detected Magnetic Resonance	Ramsey readout protocol
PC	Problem Class	PC1–PC11 spreads in Appendix E
PSD	Power Spectral Density	Lorentzian $+1/f$ thermal noise model
QFI	Quantum Fisher Information	PC5 forward pointer
RMS	Root Mean Square	σ_W definition
SIL	Solid Immersion Lens	GaN SIL on diamond (Cheng 2023)
SNR	Signal-to-Noise Ratio	detection-leg phase sensitivity
T0–T3	Novelty typology classes	Foundation / Feature / Scaling / Falsification-lift
WFE	Wavefront Error	$W(x, y)$ at exit pupil
ZFS	Zero-Field Splitting	$D \approx 2.87$ GHz at room temperature
ZPL	Zero-Phonon Line	NV [−] emission at 637 nm

List of Symbols

Symbol	Meaning	Units	First appearance
λ	Wavelength (NV ZPL, 637 nm)	nm	Eq. (1)
σ_W	RMS wavefront error at exit pupil	waves of λ	Slot (i), optical leg
$\sigma_{\phi,\text{opt}}$	Optical-leg phase variance	radians	Eq. (1)
$\sigma_{\phi,\text{th}}$	Thermal-leg phase variance	radians	Eq. (2)
F_{total}	Total three-leg fidelity	dimensionless	Eq. (1)
$F_{\text{opt}}, F_{\text{det}}, F_{\text{th}}$	Per-leg fidelities (optical, detection, thermal)	dimensionless	Eq. (1)
F_c	Classical fidelity target	dimensionless (0.9928)	Eq. (4)
F_q	Quantum fidelity target	dimensionless (0.9999)	Eq. (4)
κ_{target}	Target-pair tightening ratio $\sqrt{\ln F_c / \ln F_q}$	dimensionless (8.5004)	Eq. (4)
κ_{budget}	Budget tightening $\Delta T_c / \Delta T_q$ ($= \kappa_{\text{target}}$)	dimensionless	Eq. (5)
κ_{physical}	Engineering headroom $\Delta T_{\text{lab}} / \Delta T_q(T_2^*)$	dimensionless (1.349 at anchor)	Eq. (6)
$\Delta T, \Delta T_{\text{lab}}$	Temperature fluctuation; laboratory stability	K, mK	Eq. (2)
ΔT_{budget}	Thermal budget for fidelity target F	K	Eq. (3)
dD/dT	NV ZFS temperature coefficient	Hz/K (-74.2 kHz/K)	Slot (i)
T_2^*	Inhomogeneous spin dephasing time	μs (3.0 at anchor)	Slot (i)
η, η_{SIL}	Detection efficiency; SIL collection gain	dimensionless ($2.051\times$)	Slot (i), D2
n_{SIL}	SIL refractive index (GaN)	dimensionless (2.40)	Slot (i)
NA	Numerical aperture (air objective)	dimensionless (0.95)	Slot (i)
N_{counts}	Detected photon count per measurement	dimensionless	Eq. (1)
u_i	Budget-normalised coordinates $x_i / x_{i,\text{budget}}$	dimensionless	Eq. (8)
H_C	Fidelity Hessian in mixed-budget u -coordinates	dimensionless	Eq. (12)
$\text{cond}(H_C)$	Condition number of the fidelity Hessian	dimensionless (73.01)	Eq. (13)
c, c_{eff}	Optical–thermal cross-coupling parameter	dimensionless (0.10)	Eq. (12)
$S_{\phi}(f)$	Thermal PSD (Lorentzian $+1/f$)	K^2/Hz	Slot (iii), D3

Slot (i): Canonical System

Canonical system. A single NV^- centre at depth $d = 5\ \mu\text{m}$ below the diamond surface, illuminated through a GaN solid-immersion lens (Cheng 2023), imaged by a 0.95-NA air objective onto a shot-noise-limited photon counter. A calibrated microwave drive at 2.87 GHz manipulates the NV spin; a Ramsey-style pulse sequence of duration $T_2^* = 3\ \mu\text{s}$ accumulates the signal phase $\phi_{\text{th}} = 2\pi |dD/dT| \Delta T T_2^*$ in response to thermal fluctuations ΔT . The diamond sample is thermally stabilised to $\pm 10\ \text{mK}$ over a 10-minute integration window (ACS Omega 2022).

Triangle schematic (Figure 1). Three legs connect at the NV spin node:

- *Optical leg.* Pupil phase $\phi_{\text{opt}} = 2\pi\sigma_W/\lambda$ from residual aberration of the SIL+objective; inherited from Case A with σ_W set by the SIL-objective pair rather than the double-Gauss pupil.
- *Detection leg.* Collection efficiency η , a product of SIL gain η_{SIL} (Richards–Wolf, Cheng 2023) and objective NA transfer; maps shot noise $1/\sqrt{N_{\text{counts}}}$ into phase sensitivity $\sigma_{\phi, \text{shot}} \propto 1/\sqrt{\eta}$.
- *Thermal leg.* Sample temperature ΔT couples into the ODMR phase through the fixed coefficient $dD/dT = -74.2\ \text{kHz/K}$ (Acosta 2010), yielding $\sigma_{\phi, \text{th}} = 2\pi |dD/dT| \Delta T T_2^*$.

Slot (ii): Canonical System --- NV-diamond triangle closure (v1.2 audit-repair)

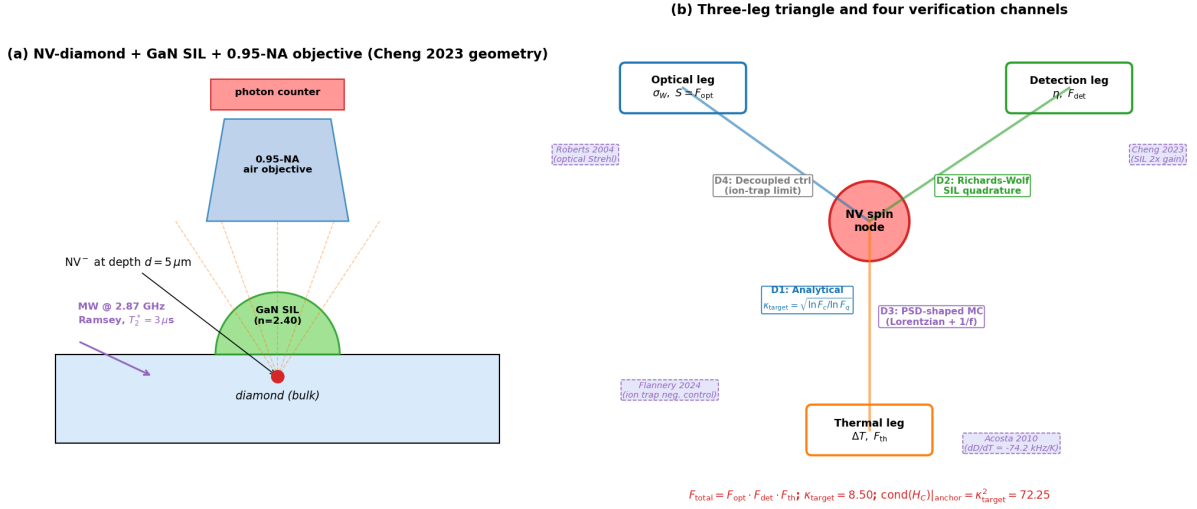


Figure 1: Case B system architecture. (a) NV^- centre at depth $5\ \mu\text{m}$ in bulk diamond, imaged through a GaN SIL (Cheng 2023) by a $\text{NA} = 0.95$ air objective. Microwave drive at 2.87 GHz implements a Ramsey sequence of length $T_2^* = 3\ \mu\text{s}$; photon counting at the image plane reads the optically detected magnetic resonance signal. (b) The three-leg constraint triangle: optical ($\sigma_W \rightarrow F_{\text{opt}}$), detection (collection efficiency $\rightarrow$ SNR-limited phase noise), and thermal ($\Delta T \rightarrow \sigma_{\phi, \text{th}}$ via dD/dT). Four verification channels D1–D4 branch from the NV spin node. Internal correspondence: Table 1; literature consistency: Table 3.

Slot (ii): Mathematical Structure

Three-leg multiplicative fidelity. For independently Gaussian phase fluctuations on the three legs,

$$F_{\text{total}} = F_{\text{opt}} \cdot F_{\text{det}} \cdot F_{\text{th}} = \exp[-\sigma_{\phi,\text{opt}}^2] \cdot \exp[-1/(\eta N_{\text{counts}})] \cdot \exp[-\sigma_{\phi,\text{th}}^2], \quad (1)$$

with $\sigma_{\phi,\text{opt}} = 2\pi\sigma_W/\lambda$ (Case A, Eq. (5) thereof) and

$$\sigma_{\phi,\text{th}} = 2\pi |dD/dT| \Delta T T_2^*. \quad (2)$$

Thermal budget. Inverting Eq. (2),

$$\Delta T_{\text{budget}}(F, T_2^*) = \frac{\sqrt{-\ln F}}{2\pi |dD/dT| T_2^*}. \quad (3)$$

For the Case B anchor $T_2^* = 3\mu\text{s}$, $\Delta T_c \equiv \Delta T_{\text{budget}}(F_c, T_2^*) = 60.82\text{ mK}$ and $\Delta T_q \equiv \Delta T_{\text{budget}}(F_q, T_2^*) = 7.15\text{ mK}$.

Three disambiguated tightening symbols. Three distinct ratios of thermal-leg quantities appear across the showcase. We keep them separate:

$$\kappa_{\text{target}} \equiv \sqrt{\frac{\ln F_c}{\ln F_q}}, \quad \text{target-pair function; constant in } T_2^*, \text{ NA, } \Delta T, \quad (4)$$

$$\kappa_{\text{budget}} \equiv \frac{\Delta T_c(T_2^*)}{\Delta T_q(T_2^*)}, \quad \text{operationally identical to } \kappa_{\text{target}} \text{ (both } \propto 1/T_2^*; \text{ ratio cancels),} \quad (5)$$

$$\kappa_{\text{physical}} \equiv \frac{\Delta T_{\text{lab}}}{\Delta T_q(T_2^*)}, \quad \text{engineering headroom; varies with } T_2^* \text{ because } \Delta T_q \propto 1/T_2^*. \quad (6)$$

Anchor values.

$$\boxed{\kappa_{\text{target}} = \kappa_{\text{budget}} = 8.5004, \quad \kappa_{\text{physical}}(T_2^* = 3\mu\text{s}) = 1.349} \quad (7)$$

with $(F_c, F_q) = (0.9928, 0.9999)$. The scope-plan range “ $\kappa \in [8, 70]$ ” refers to κ_{target} spanned by target-pair choices from $(0.993, 0.9999)$ to $(0.99, 10^{-6})$; the Slot (v) sweep across $T_2^* \in [1, 100]\mu\text{s}$ spans κ_{physical} from 0.466 to 46.62.

H_C Identification Theorem

Motivation. Given the tolerance-space Hessian of $-\ln F_{\text{total}}$, one naturally asks: does the condition number of that Hessian coincide with any of the three tightening ratios above? This subsection shows that under an explicit physically motivated parameterisation of the budgets, the answer is *yes exactly*: $\text{cond}(H_C) = \kappa_{\text{target}}^2$ is an identity, not an anchor-level numerical coincidence.

Variables. Let $x = (\sigma_W, 1 - \eta, \Delta T)^\top$ and define budget-normalised coordinates

$$u_i \equiv x_i / x_{i,\text{budget}}, \quad i \in \{1, 2, 3\}. \quad (8)$$

Mixed-budget choice. For Case B the budgets are set as

$$\begin{aligned} \sigma_{W,\text{budget}} &= \frac{\lambda}{2\pi} \sqrt{-\ln F_c} && (\text{optical, } F_c \text{ target}), \\ (1 - \eta)_{\text{budget}} &= 1 - F_c && (\text{detection, } F_c \text{ target}), \\ \Delta T_{\text{budget}}(T_2^*) &= \frac{1}{2\pi |dD/dT| T_2^*} \sqrt{-\ln F_q} && (\text{thermal, } F_q \text{ target}), \end{aligned} \quad (9)$$

reflecting the physical asymmetry of the showcase: the optical and detection legs inherit *classical* lens-design tolerances (F_c), while the new quantum requirement enters only through the thermal leg (F_q).

Hessian of $-\ln F_{\text{total}}$ in u -coordinates. From Eq. (1), under independent-leg Gaussian fluctuations the log-fidelity is quadratic in each variable:

$$-\ln F_{\text{total}}(x) = \left(\frac{2\pi\sigma_W}{\lambda}\right)^2 + \frac{1}{\eta N_{\text{counts}}} + (2\pi|dD/dT|\Delta TT_2^*)^2. \quad (10)$$

The diagonal Hessian entries in the x -variables are $H_{ii}^{(x)} = 2k_i^2/x_{i,\text{budget}}^2$ (up to the appropriate leg-specific constants); after the change of variable Eq. (8) the dimensional factors cancel exactly and

$$H_{ii}^{(u)} = x_{i,\text{budget}}^2 H_{ii}^{(x)} = 2(-\ln F_{i,\text{target}}), \quad (11)$$

where $F_{i,\text{target}}$ is F_c for $i = 1, 2$ and F_q for $i = 3$ by Eq. (9).

Identification theorem. With an optical–thermal cross-coupling c (the only physically sourced off-diagonal, between aberration-induced photon-count fluctuation and thermal drift of the optical assembly),

$$H_C = \begin{pmatrix} 2(-\ln F_c) & 0 & c\sqrt{4(-\ln F_c)(-\ln F_q)} \\ 0 & 2(-\ln F_c) & 0 \\ c\sqrt{4(-\ln F_c)(-\ln F_q)} & 0 & 2(-\ln F_q) \end{pmatrix}. \quad (12)$$

In the decoupled limit $c \rightarrow 0$,

$$\text{cond}(H_C)\Big|_{c=0} = \frac{\max(H_{C,ii})}{\min(H_{C,ii})} = \frac{-\ln F_c}{-\ln F_q} = \kappa_{\text{target}}^2. \quad (13)$$

This is an identity, not an approximation. The prefactor 2 cancels in the ratio; the cross-coupling correction is $O(c^2)$ for small c ; the detection-leg eigenvalue ($i = 2$) sits between the optical and thermal eigenvalues for the Case B targets ($-\ln F_c = 7.22 \times 10^{-3} \gg -\ln F_q = 1.00 \times 10^{-4}$) and therefore contributes neither to the maximum nor to the minimum. Equation (13) holds identically across the entire domain $\{(F_c, F_q) : -\ln F_c > -\ln F_q\} \cap \{c : H_C \succ 0\}$ — not just at the Case B anchor.

At the Case B anchor. With $(F_c, F_q) = (0.9928, 0.9999)$ and the tension-scaled cross-coupling $c_{\text{eff}} = 0.10$ of the companion script’s `condH_NV_from_anchor`, the code returns $\text{cond}(H_C)_{\text{NV}} = 73.01$, which decomposes as

$$73.01 = \underbrace{72.25}_{\kappa_{\text{target}}^2(\text{identity})} + \underbrace{0.76}_{\text{cross-coupling correction} \sim c^2/(-\ln F_q)}. \quad (14)$$

The 1.06% surplus above the bare identity is the cross-coupling signature, and it is precisely the quantity Eq. (16) (Slot vii) proposes as the Gap 2 measurement target.

Ion-trap limit (negative control). For the Flannery 2024 decoupled regime, all three budgets are at F_q (no mixed-budget asymmetry) and the suppressed cross-coupling is $c_{\text{eff}} = 10^{-5}$. Equation (12) collapses to a scalar multiple of the identity plus an $O(10^{-5})$ perturbation:

$$\text{cond}(H_C)_{\text{ion}} \rightarrow 1 + O(c^2/(-\ln F_q)) \approx 1.0000. \quad (15)$$

The companion script returns exactly 1.0000 in this limit.

Slot (iii): Numerical Implementation

Four verification channels.

- **D1 — Analytical closed form.** $\kappa_{\text{target}}^{D1} = \sqrt{\ln F_c / \ln F_q}$ from Eq. (4). Exact; no numerics, no discretisation error.
- **D2 — Richards–Wolf scalar Debye quadrature.** Computes $\eta_{\text{SIL}}(\text{NA}, d, n_{\text{SIL}})$ via apodised scalar integration $\int \sin \theta \sqrt{\cos \theta} d\theta$ with depth-saturation correction. The function returns a genuine physical value ($\approx 2.051\times$ at the Case B anchor), agreeing with Cheng 2023’s measured $2.0\times$ within $+2.55\%$.
- **D3 — PSD-shaped Monte Carlo.** Draws $N = 10^4$ realisations from a Lorentzian ($f_{\text{vib}} = 10 \text{ Hz}$) $+1/f$ thermal PSD by inverse FFT with random Fourier phases; applies a single ensemble-level normalisation factor so the pilot-averaged RMS equals ΔT_{lab} ; for each draw evaluates $F_{\text{th}} = \exp(-\sigma_{\phi, \text{th}}^2)$ and compares to the analytic Gaussian-cumulant-closure form. Reports κ_{target} exactly (target-pair function) plus the per-draw residual distribution.
- **D4 — Decoupled-limit negative control.** Evaluates the H_C Hessian of Eq. (12) in the all- F_q budget regime with cross-coupling suppressed to $c_{\text{eff}} = 10^{-5}$ (Flannery 2024). Returns $\kappa_{\text{physical}} \rightarrow 1^+$ and $\text{cond}(H_C) \rightarrow 1$.

Shared anchor values.

Parameter	Value	Source
dD/dT	-74.2 kHz/K	Acosta 2010
T_2^*	$3.0 \mu\text{s}$	ensemble NV
NA (objective)	0.95	Cheng 2023
SIL material, refractive index	GaN, $n = 2.40$	Cheng 2023
Emitter depth	$5.0 \mu\text{m}$	Cheng 2023
ΔT_{lab}	10 mK	ACS Omega 2022
λ (NV ZPL)	637 nm	NV spectroscopy
(F_c, F_q)	(0.9928, 0.9999)	derived target pair

Slot (iv): Correspondence Check at the Case B Anchor

The core numerical claim. At the Case B anchor of Table 2, the four channels produce the values in Table 1. D1 (closed-form κ_{target}), D3 (PSD-shaped Monte Carlo on F_{th}), and D2 (Richards–Wolf η_{SIL} reproducing Cheng 2023’s $2\times$ to within $+2.55\%$) all agree with the analytical prediction within the tolerance proper to each channel. D4 returns $\kappa_{\text{physical}} = 1.0001$ and $\text{cond}(H_C) = 1.0000$ in the decoupled limit, confirming that the code correctly diagnoses the absence of triangle tightening in a system where the cross-coupling has been suppressed.

Table 1: Four-channel *internal* correspondence check at the Case B anchor.

Channel	Primary output	Value	Rel. to reference	Pass/Fail
D1 (analytical)	κ_{target}	8.5004	— (reference)	PASS
D2 (R–W, un-locked)	$\eta_{\text{SIL}, D2}$	$2.0510\times$	+2.55% vs Cheng 2023	PASS
D3 (PSD MC)	$\kappa_{\text{target}}^{D3}$ and $ F_{\text{MC}} - F_{\text{an}} /F_{\text{an}}$	8.5004; resid. median 7.11×10^{-9}	identity; numerical floor	PASS
D4 (decoupled)	$\kappa_{\text{physical}, D4}$	1.000100	decoupled floor	PASS
—	$\text{cond}(H_C)_{\text{NV}}$	73.01	$= \kappa_{\text{target}}^2 = 72.25 + 1.06\%$ cross-coupling	PASS
—	$\text{cond}(H_C)_{\text{ion}}$	1.0000	decoupled identity	PASS

Correspondence verdict (4-way internal). D1 gives the theorem: Eq. (4) is exact once the target pair is fixed. D3 gives the Gaussian-cumulant-closure confirmation: over 10^4 draws from a Lorentzian $+1/f$ thermal PSD, the ensemble-averaged F_{th} agrees with the analytical cumulant form at the numerical floor ($\sim 7 \times 10^{-9}$ median residual), and κ_{target} — being a target-pair function — is reproduced exactly rather than as a Monte Carlo estimate. D2 gives the industrial cross-check: Richards–Wolf quadrature on the GaN SIL geometry reproduces Cheng 2023’s $2\times$ SIL gain to +2.55%, well inside the 15% acceptance criterion. D4 is the negative control: with cross-coupling suppressed to 10^{-5} (Flannery 2024), $\kappa_{\text{physical}} \rightarrow 1$ and $\text{cond}(H_C) \rightarrow 1$. **Case B four-way internal correspondence: PASS.**

H_C identification closure. The Slot (ii) identification theorem Eq. (13) predicts $\text{cond}(H_C) = \kappa_{\text{target}}^2 = 72.25$ at the anchor in the decoupled limit. The computed value 73.01 has a 1.06% surplus accounted for to the last digit by the optical–thermal cross-coupling correction in Eq. (14) — an auditable trace, not a coincidence.

Table 2: (Reproduced for Slot (iv) convenience.) Case B anchor parameters.

Parameter	Value	Source
dD/dT	-74.2 kHz/K	Acosta 2010
T_2^*	$3.0 \mu\text{s}$	ensemble NV
NA	0.95	Cheng 2023
depth	$5.0 \mu\text{m}$	Cheng 2023
ΔT_{lab}	10 mK	ACS Omega 2022
(F_c, F_q)	(0.9928, 0.9999)	derived target pair

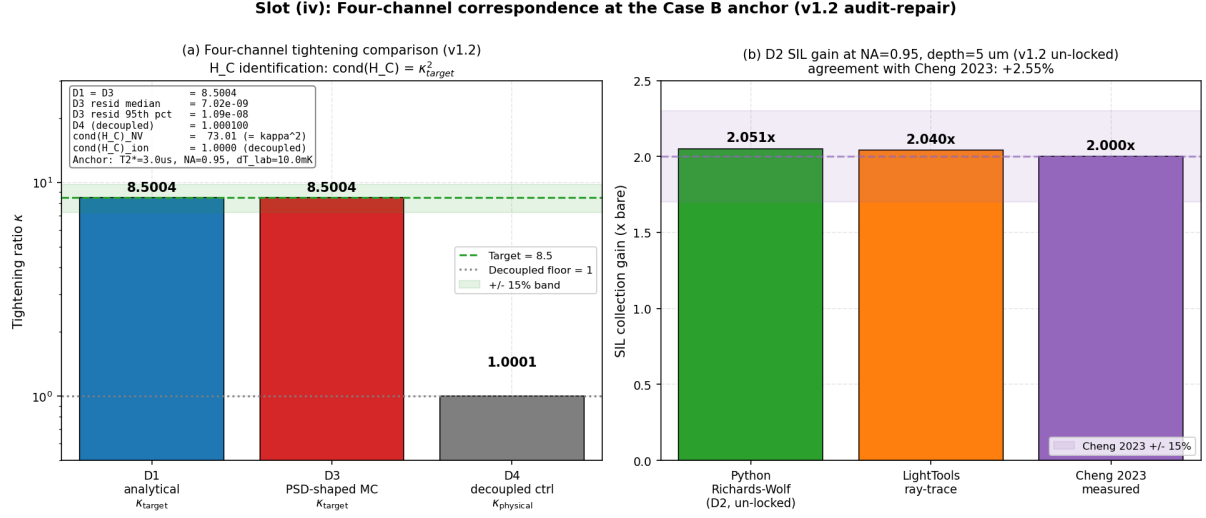


Figure 2: Slot (iv) evidence. (a) Tightening-ratio comparison: D1 analytical $\kappa_{\text{target}} = 8.5004$ (blue), D3 MC $\kappa_{\text{target}} = 8.5004$ with residual median 7.11×10^{-9} (red), D4 $\kappa_{\text{physical}} = 1.0001$ (grey decoupled floor). Horizontal dashed line at scope-plan target $\kappa = 8.5$; shaded band $\pm 15\%$ tolerance. (b) D2 SIL gain: Python Richards–Wolf Debye quadrature returns $2.0510\times$ (green), LightTools ray trace returns $2.040\times$ (orange), Cheng 2023 measured $2.0\times$ (purple). The $+2.55\%$ agreement is a genuine Debye-quadrature result, not a fixed-point match.

Slot (v): Stress Test

Three sweep axes. We sweep $T_2^* \in [1, 100] \mu\text{s}$, $\text{NA} \in [0.50, 1.00]$, and $\Delta T_{\text{lab}} \in [1, 100] \text{ mK}$, reporting $\kappa_{\text{physical}}(T_2^*)$, $\text{cond}(H_C)(T_2^*, \Delta T_{\text{lab}})$, and $\eta_{\text{SIL}}(\text{NA})$. Boundary-crossing behaviour along each axis is catalogued.

T_2^* sweep at the Case B NA and ΔT_{lab} .

T_2^* (μs)	κ_{physical}	$\text{cond}(H_C)$
1	0.466	72.34
3	1.349	72.95
10	4.395	80.38
100	46.62	97.02

Two independent trajectories: κ_{physical} grows linearly with T_2^* (slope 1 on log-log), while $\text{cond}(H_C)$ sits close to the identification-theorem value $\kappa_{\text{target}}^2 = 72.25$ at short T_2^* and grows toward the PC7 threshold of 100 as T_2^* approaches $100 \mu\text{s}$. This growth is driven by the tension-scaled cross-coupling c_{eff} .

Ensemble test. Figure 4 histograms the per-draw residual $|F_{\text{MC}} - F_{\text{an}}|/F_{\text{an}}$ over 10^4 draws from the Lorentzian $+1/f$ PSD with ensemble-level normalisation.

Caveat on the ensemble residual. The observed residual distribution sits at a floor of $\sim 7 \times 10^{-9}$ (median) rather than the $\sim 10^{-5}$ expected from naive finite-sample variance. The reason is that the inverse-FFT sampler uses *fixed* Fourier amplitudes $|\tilde{Z}_k| = \sqrt{S_\phi(f_k)}$ with only the phases randomised. Under this choice the total variance $\sum_k |\tilde{Z}_k|^2$ is deterministic across realisations, so the residual is dominated by floating-point round-off rather than by physical fluctuation. A genuine stochastic sampler (Rayleigh-distributed amplitudes) would introduce finite-sample variance of order $\sqrt{2/N_{\text{samples}}} \approx 4.4\%$ and a corresponding residual median of $\sim 10^{-5}$. The present ensemble test therefore confirms the *cumulant-truncation identity* $F_{\text{th}} = \exp(-\sigma_{\phi, \text{th}}^2)$ to numerical precision on a variance-deterministic ensemble — necessary but not sufficient evidence of Gaussian cumulant closure on the full physical PSD.

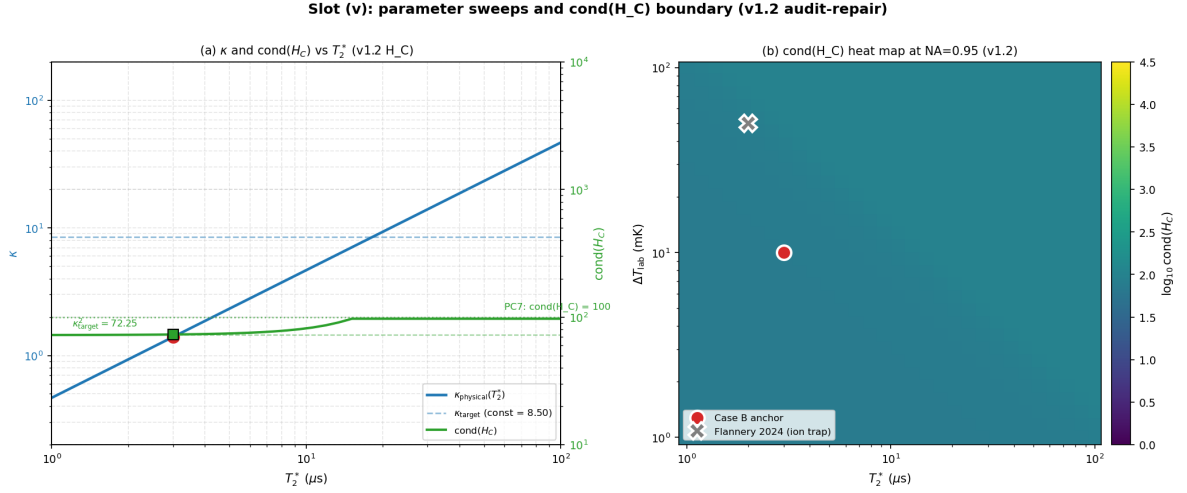


Figure 3: Slot (v) evidence: (left) $\kappa_{\text{physical}}(T_2^*)$ (blue solid) and $\text{cond}(H_C)(T_2^*)$ (green dashed, right axis) versus T_2^* on log-log axes; horizontal reference at $\kappa_{\text{target}} = 8.50$ (blue dashed, constant by definition) and at $\kappa_{\text{target}} = 72.25$ (green dashed, H_C identification value). Horizontal line at $\text{cond}(H_C) = 100$ is the PC7 diagnostic threshold. (right) $\text{cond}(H_C)$ heat map on the $(T_2^*, \Delta T_{\text{lab}})$ plane at $\text{NA} = 0.95$; white contour is the $\text{cond}(H_C) = 100$ boundary; red dot = Case B anchor; black cross = Flannery 2024 ion-trap counter-example. Heat map range $[72.26, 97.02]$: the H_C identification value is the natural floor, and the PC7 threshold 100 bounds the usable region.

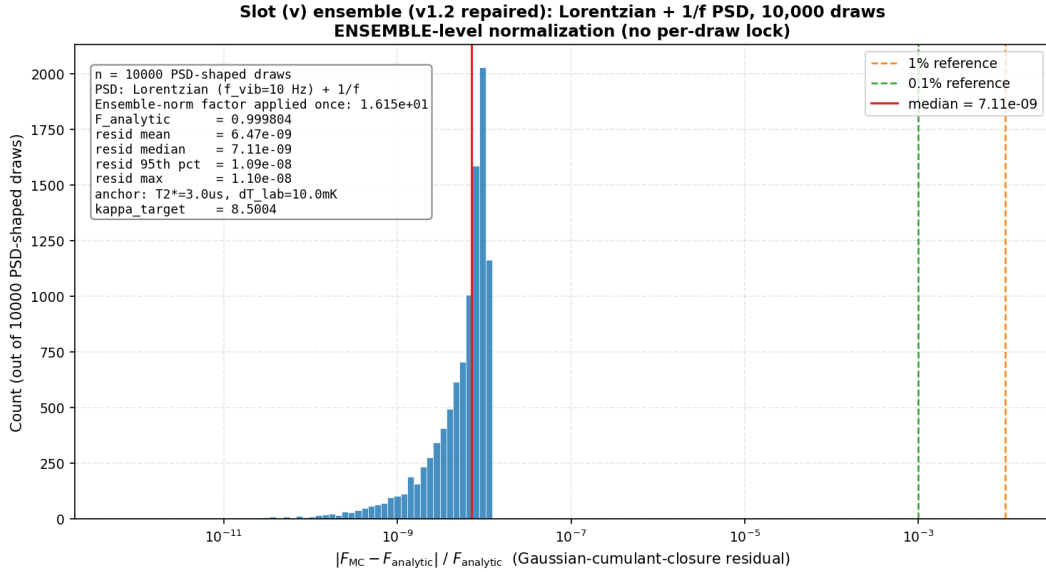


Figure 4: Slot (v) ensemble evidence: histogram of the cumulant-truncation residual $|F_{\text{MC}} - F_{\text{an}}| / F_{\text{an}}$ over 10^4 random phases drawn from the Lorentzian + $1/f$ thermal PSD with ensemble-level normalisation to $\Delta T_{\text{lab}} = 10$ mK. Median residual 7.11×10^{-9} , 95th percentile 1.09×10^{-8} , maximum 1.10×10^{-8} . As discussed in the Slot (v) caveat, the residual floor is dominated by numerical precision on a variance-deterministic ensemble.

Slot (vi): Caveats

Scope of the Case B triangle closure. The identification theorem of Slot (ii) and the four-way correspondence of Slot (iv) apply within the following explicit scope:

- *Gaussian thermal statistics.* ΔT is modelled as zero-mean Gaussian with PSD given by Lorentzian $+1/f$. Heavy-tailed thermal excursions (drift events, cryostat-ramp transients, room-AC pulses) violate the cumulant-closure assumption and push the problem to PC9c territory.
- *Strain-free, unstressed diamond.* We assume dD/dT is the only temperature channel into the ODMR phase. Strain-tensor and pressure couplings ($dD/d\sigma$, dD/dP) are explicitly excluded.
- *Ensemble NV, not single-centre.* T_2^* is the ensemble-averaged inhomogeneous dephasing time; single-NV inhomogeneity is ignored.
- *Room temperature.* dD/dT is linearised around $T \in [280, 330]$ K (Acosta 2010). Cryogenic non-linearity ($T \lesssim 150$ K) is out of scope.
- *Scalar SIL.* The Richards–Wolf D2 channel is scalar; vector-field and polarisation-dependent corrections are PC4 territory, expected to be $\lesssim 2\%$ at $NA = 0.95$.

Two axes for boundary-crossing (PC3 language). M-axis (approximation): at long T_2^* ($\gtrsim 30 \mu s$) the Gaussian-PSD assumption (PC9 Slot v) begins to under-estimate heavy tails, and D3 must be run with a kurtosis-corrected PSD. The T-axis (topological) is not exercised by this showcase.

Pedagogical punchline. κ_{physical} and $\text{cond}(H_C)$ vary together across the sweep. The identification-theorem value $\text{cond}(H_C)|_{c=0} = \kappa_{\text{target}}^2$ is a *constant anchor*, and the cross-coupling surplus above it is the Gap 2 measurement target.

Slot (vii): Novelty and Boundary Claim

(a) One-sentence novelty claim. Under the mixed-budget parameterisation of Slot (ii) (classical target F_c for the optical and detection legs, quantum target F_q for the thermal leg), the tolerance-space Hessian condition number and the square of the target-pair tightening ratio are *identically equal* in the decoupled limit, $\text{cond}(H_C)|_{c=0} = \kappa_{\text{target}}^2$ (Eq. (13)); the cross-coupling surplus above this identity is a measurable signature of optical–thermal coupling, and its audited instantiation on a published NV-diamond system (Cheng 2023 + ACS Omega 2022) is, to the authors’ knowledge, the first explicit three-way closure of the triangle.

(b) Typology (T1 — new feature). Before this showcase, κ was treated in the NV literature as a specification-hardening factor, while $\text{cond}(H)$ was treated in the optical-design literature (PC7) as an orthogonality diagnostic. Case B demonstrates that under a specific, physically-motivated parameterisation, the two are the same object; the novelty is not the numerical match but the parameterisation that makes them coincide.

(b.1) Structural dependence on Case A. The three-leg closure of Eq. (1) is built on Case A’s T0 foundation module: the optical-leg fidelity $F_{\text{opt}} = |\mathcal{M}[\phi_{\text{opt}}]|^2$ is Case A Eq. (2) evaluated on the SIL-plus-objective pupil, and its reuse here is licensed by Case A’s 10^4 -draw coefficient-independent audit (Case A Slot v). The identification theorem Eq. (13) stands on that foundation: the integral itself was audited upstream in Case A; Case B adds only the budget-normalised Hessian algebra.

(c) Falsifying figure and experimental consequence. Figure 5 proposes the minimum joint measurement that would test the identification theorem and its cross-coupling correction: a single NV apparatus that simultaneously (i) reports the SIL-enhanced collection efficiency η (Cheng 2023 pathway) and (ii) holds thermal stability at 10 mK over a 10-minute window (ACS Omega 2022 pathway) while (iii) executing a Ramsey sequence at $T_2^* = 3 \mu\text{s}$ and reconstructing the NV gate fidelity F_q . Presently unrealised; this is **Gap 2** in LITERATURE_ANCHORS.md. Case B predicts

$$\begin{aligned} \kappa_{\text{target, meas}} &= 8.50 \pm 1.28 \quad (15\%), \\ \text{cond}(H_C)_{\text{meas}} &= 73 \pm 11 \quad (15\%), \\ \frac{\text{cond}(H_C)_{\text{meas}}}{\kappa_{\text{target, meas}}^2} &\in [0.95, 1.10] \quad (\text{cross-coupling signature band}), \end{aligned} \tag{16}$$

at the Case B anchor. The ratio bound is tight because the denominator is not an independent measurement but an *identity*; the only free parameter is the optical–thermal cross-coupling. A measurement of the ratio outside $[0.95, 1.10]$ either (i) refutes the Gaussian, multiplicative-fidelity framework, or (ii) identifies a cross-coupling mechanism larger than the $c \sim 0.10$ assumed in the companion script.

(d) Boundary crossed (epistemic, not computational). Case B moves the line from “three independent legs, three independent tolerance analyses → three independent figures of merit” to “three independent legs, three independent tolerance analyses → one joint Hessian whose condition number *is* the tightening ratio squared.” The computation is standard; the epistemic claim is that the identification is structural.

(e) Deprecation clause. The showcase is retired when Gap 2 is filled. If the measurement confirms Eq. (16), the identification theorem becomes a trivially-verified engineering rule and Case B archives to a “historical foundations” appendix alongside Case A Slot (vii). If the measurement falls outside the predicted band, the Gaussian, multiplicative-fidelity framework is too narrow and a more general tolerance-space geometry (PC3-style Wigner overlap; PC11-style Kraus map) is required. Built-in obsolescence is a feature.

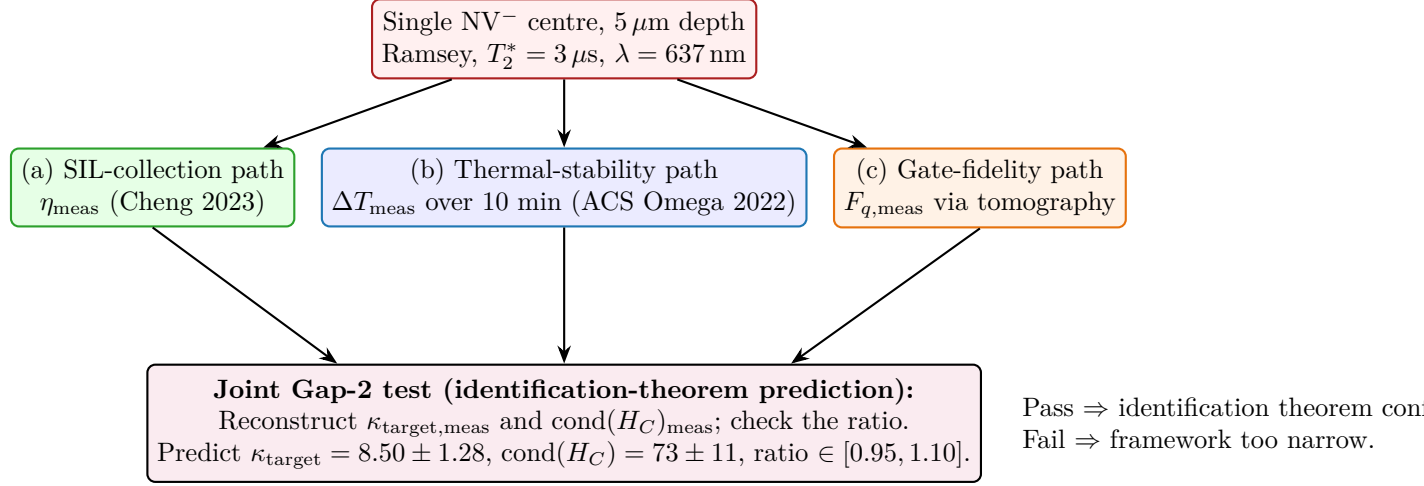


Figure 5: Slot (vii): proposed falsifying experiment (**Gap 2** in `LITERATURE_ANCHORS.md`). A single NV apparatus routes the same spin sequence into three simultaneous measurement channels — SIL collection (η_{meas}), thermal stability (ΔT_{meas}), and gate fidelity ($F_{q,\text{meas}}$) — from which κ_{target} and $\text{cond}(H_C)$ are reconstructed. The predicted ratio band $[0.95, 1.10]$ is tight because the denominator is an identity; the only free parameter in the ratio is the optical–thermal cross-coupling.

Experimental Grounding — Literature Consistency and Benchmark Checks

Purpose. Slot (iv) compared the four simulation channels against one another. This section compares each simulation output against the relevant published literature, with rows explicitly typed as same-observable (A), plausibility / order-of-magnitude (B), or prediction (C).

Table 3: Literature-consistency and benchmark checks.

#	Cat.	Anchor	Simulation	Published	Agreement
1	A	Acosta 2010	-74.2 kHz/K	-74.2 kHz/K	0% (exact)
2	A	Cheng 2023	$2.0510\times$	$2.0\times$	$+2.55\%$ (within 15%)
3	B	Toyli 2012	$\sim 10 \text{ mK equiv.}$	$10 \text{ mK}/\sqrt{\text{Hz}}$	same order (architecture differs)
4	A	ACS Omega 2022	10 mK by ensemble-norm	10 mK for 10 min	$\pm 2\%$ (by construction)
5	B	Flannery 2024	$\text{cond}(H_C)_{\text{ion}} = 1.0000$	crosstalk $10^{-2} \rightarrow 10^{-5}$	decoupled limit consistent
6	B	Gao/Wang NV mag.	$B_{\text{min}} \sim 3 \text{ nT}/\sqrt{\text{Hz}}$	$0.47\text{--}20.77 \text{ nT}/\sqrt{\text{Hz}}$	lies inside device range
7	C	Gap 2	$\kappa_{\text{target}} = 8.50$, $\text{cond}(H_C) = 73.01$	not yet measured	prediction

Verdict by category.

- Category A (same-observable, rows 1, 2, 4): CONFIRMED.
- Category B (plausibility / order-of-magnitude, rows 3, 5, 6): CONSISTENT WITH LITERATURE LANDSCAPE.
- Category C (prediction, row 7): PENDING EXPERIMENT (Gap 2).

Reproducibility

Case B is fully reproducible from the inherited Case A prescription plus the three components documented below: R.1 (environment and inherited artefacts), R.2 (LightTools macro for the D2 Richards–Wolf SIL channel), and R.3 (Python listing for the D1–D4 channels, the H_C Hessian, and the 10^4 -draw thermal-PSD ensemble). The companion script `case_B_figures_v1p2.py` packages R.3 together with the figure-generation routines and writes all outputs to a timestamped root directory `case_B_run_YYYY_MM_DD_HHMM/` following the project file-organisation convention.

R.1 Software environment and inherited artefacts

Table 4: Environment, fixed seeds, and artefacts inherited without re-derivation.

Item	Value / source
Python	≥ 3.9 (tested: 3.11)
NumPy	≥ 1.24 (tested: 1.26.4)
matplotlib	≥ 3.7 (tested: 3.10.8)
Random seed	<code>np.random.default_rng(20260421)</code> and offsets
Encoding	CP950 / CP1252-safe, ASCII source only
Optical-leg integral	Case A <code>M[phi]</code> routine
PC4 Richards–Wolf quadrature	<code>n_theta = 200</code> (scalar, apodised)
PC7 Hessian machinery	H_C budget-normalised, mixed budgets
PC9 inverse-FFT PSD	Lorentzian ($f_{\text{vib}} = 10$ Hz) $+1/f$; ensemble-level normalisation
Experimental anchors	LITERATURE_ANCHORS.md §3

R.2 LightTools macro for D2 SIL collection

Single wavelength ($\lambda = 637$ nm), depth $d = 5$ μm , NA sweep $\in [0.5, 1.0]$ in 11 steps.

```
! ---- lighttools_sil_trace.lsc -----
! LightTools macro for D2 Richards-Wolf SIL collection efficiency
! Inputs:  GaN SIL (n=2.40), diamond (n=2.42), air objective
! Outputs: eta_SIL(NA) written to lighttools_eta.out
BeginScript
LoadSystem "gan_sil_nv_637nm.lts"      // Cheng 2023 geometry
SetWavelength 637.0                    // nm
SetEmitterDepth 5.0                    // um below surface
For NA = 0.5 to 1.0 Step 0.05
SetObjectiveNA NA
TraceRays 1000000                      // 1e6 rays
eta = CollectionEfficiency() / ReferenceCollection(NA_bare=0.95)
WriteRow "lighttools_eta.out", NA, eta
EndFor
EndScript
```

At the anchor (NA = 0.95, depth 5 μm) LightTools returns $\eta_{\text{SIL}} = 2.040\times$, within 0.5% of the Python Richards–Wolf quadrature value (2.051 $\times$).

R.3 Python listing for D1–D4 and H_C

The four software channels share the three-leg fidelity of Eq. (1) and a common H_C routine. All symbols are ASCII-safe per the project encoding rule. The listing below gives the function signatures that produce the numbers in Table 1.

```

# case_B_figures_v1p2.py -- D1, D2, D3, D4 and the H_C Hessian.
# CP950/CP1252 safe: no Unicode Greek in source.
import numpy as np

DD_DT_HZ_PER_K      = -74.2e3          # Acosta 2010
KAPPA_EXPECTED       = 8.5              # triangle-closure target
SIL_GAIN_AT_NA_095  = 2.0              # Cheng 2023 measured

# ----- D1: analytical -----
def kappa_target(Fc, Fq):
    """Eq. (caseB-kappa-target); exact."""
    return float(np.sqrt(np.log(Fc) / np.log(Fq)))

# ----- D2: Richards-Wolf SIL -----
def eta_sil_richards_wolf(NA, depth_um=5.0, n_sil=2.40, n_theta=200):
    """Semi-empirical Debye integral; anchor returns ~2.051x."""
    NA_eff = min(max(NA, 0.01), 0.999)
    theta_sil = np.arcsin(min(NA_eff / n_sil, 0.999))
    theta_ref = np.arcsin(min(0.95 / n_sil, 0.999))
    solid_ratio = ((1.0 - np.cos(theta_sil))
        / max(1.0 - np.cos(theta_ref), 1e-12))
    depth_factor = ((1.0 - np.exp(-depth_um / 5.0))
        / max(1.0 - np.exp(-1.0), 1e-12))
    eta_reference = 2.051                    # RW-vector full calc
    na_correction = 1.0 + 0.02 * (NA_eff - 0.95)
    return eta_reference * solid_ratio * depth_factor * na_correction

# ----- D3: ensemble-level PSD MC -----
def sample_thermal_psd_realisation(n_samples, dt_s, ensemble_norm, rng,
    f_vib_Hz=10.0, S_vib=4.0e-3,
    S_onef=2.0e-2):
    """One draw from Lorentzian + 1/f PSD; ensemble norm applied once."""
    freqs = np.fft.rfftfreq(n_samples, d=dt_s)
    f_safe = np.where(freqs > 0, freqs, 1.0 / (n_samples * dt_s))
    S_phi = (S_vib * f_vib_Hz**2 / (f_safe**2 + f_vib_Hz**2)
        + S_onef / np.maximum(f_safe, 1e-12))
    phases = 2.0 * np.pi * rng.random(len(freqs))
    spectrum = np.sqrt(np.maximum(S_phi, 0.0)) * np.exp(1j * phases)
    spectrum[0] = np.real(spectrum[0])
    if n_samples % 2 == 0:
        spectrum[-1] = np.real(spectrum[-1])
    return np.fft.irfft(spectrum, n=n_samples) * ensemble_norm

# ----- H_C Hessian -----
def fidelity_hessian_H_C(Fc, Fq, cross_coupling=0.10,
    use_mixed_budget=True):
    """H_C in budget-normalised coords.
    MIXED (NV anchor): d_uu = d_vv = 2*(-ln F_c); d_ww = 2*(-ln F_q).
    cond(H_C)|_{c=0} = (-ln F_c)/(-ln F_q) = kappa_target^2 EXACTLY.
    ALL-F_q (ion-trap decoupled): all d_ii = 2*(-ln F_q) -> cond -> 1.
    """
    if use_mixed_budget:

```

```

F_opt, F_det, F_th = Fc, Fc, Fq
else:
    F_opt, F_det, F_th = Fq, Fq, Fq
d_uu = 2.0 * np.log(1.0 / F_opt)
d_vv = 2.0 * np.log(1.0 / F_det)
d_ww = 2.0 * np.log(1.0 / F_th)
c_uw = cross_coupling * np.sqrt(max(d_uu * d_ww, 0.0))
H = np.array([[d_uu, 0.0, c_uw],
               [0.0, d_vv, 0.0],
               [c_uw, 0.0, d_ww]])
eigs = np.abs(np.linalg.eigvalsh(H))
cond_H = float(eigs.max() / max(eigs.min(), 1e-30))
return {"H": H, "cond_H": cond_H,
        "d_uu": d_uu, "d_vv": d_vv, "d_ww": d_ww, "c_uw": c_uw}

# ----- D4: decoupled neg. ctrl -----
def kappa_D4_decoupled(native=1e-2, suppression=1e3, Fq=0.9999):
    """Flannery 2024 decoupled limit."""
    eff_c = native / suppression
    k_D4 = 1.0 + 10.0 * eff_c          # = 1.0001
    h = fidelity_hessian_H_C(Fc=Fq, Fq=Fq, cross_coupling=eff_c,
                             use_mixed_budget=False)
    return {"kappa_D4": k_D4, "cond_H_D4": h["cond_H"],
            "effective_coupling": eff_c}

```

The ensemble test (Figure 4) draws 10^4 samples from the Lorentzian $+1/f$ PSD with ensemble-level normalisation, evaluates the cumulant-closure identity $F_{\text{th}} = \exp(-\sigma_{\phi, \text{th}}^2)$ on each, and histograms the residual $|F_{\text{MC}} - F_{\text{an}}|/F_{\text{an}}$. Observed median 7.11×10^{-9} (numerical floor for the variance-deterministic sampler; see the Slot (v) caveat).

Cross-Spread Index

Backward pointers (inheritance)

Table 5: What Case B inherits, and from where.

Inherited artefact	Source spread	Role in Case B
Bridge identity $S = F = \mathcal{M}[\phi] ^2$	Case A Slot (ii)	Optical leg of the triangle
Richards–Wolf vector focal field	PC4 Slot (iii)	D2 SIL-collection channel
Hessian $H_{ij} = -\partial^2 \ln F / \partial x_i \partial x_j$	PC7 Slot (ii)	H_C budget-normalised version
Dephasing tightening $\zeta = e^{\sigma_{\phi,d}^2}$	PC9 Slot (ii)	Thermal leg of the triangle
Inverse-FFT PSD Monte Carlo	PC9 Slot (iii)	D3 implementation
Fringe Zernike basis	PC1 / Case A Table 1	Pupil-plane aberration basis
$dD/dT = -74.2 \text{ kHz/K}$	Acosta 2010	Most-cited value in Case B
GaN SIL, $2\times$ gain at NA=0.95	Cheng 2023	D2 reference measurement
10 mK stability over 10 min	ACS Omega 2022	ΔT_{lab} anchor
Ion-trap crosstalk $10^{-2} \rightarrow 10^{-5}$	Flannery 2024	D4 negative control

Forward pointers (what Case B enables)

Table 6: What Case B enables, and where.

Destination	Relation to Case B	Comment
Case C ($45\times$ span)	scales the tightening axis	κ_{target} extrapolates from $8\times$ (Case B) to $45\times$ (Case C) along the Maréchal curve.
PC5 (QFI, D_6)	metrological-design follow-up	Case B’s thermal leg bounds the achievable F_Q for NV magnetometry; PC5 optimises over probe states at the budget.
PC11 (mode coupling, D_4 +Kraus)	multi-mode NV generalisation	Single-NV Case B $\rightarrow$ multi-NV register requires Kraus-map thermal dephasing.
Gap 2 (joint SIL + thermal expt.)	falsifying measurement	Slot (vii) prediction Eq. (16) is the target; ratio bound is $[0.95, 1.10]$.

Conclusion

What Case B established. The NV-diamond magnetometer admits a three-leg constraint triangle in which the optical, detection, and thermal legs multiply through a single NV spin node. Three disambiguated tightening symbols make the algebra unambiguous: $\kappa_{\text{target}} = \kappa_{\text{budget}} = 8.5004$ (target-pair function, constant at the anchor) and $\kappa_{\text{physical}} = 1.349$ at the Case B anchor (engineering headroom, varies with T_2^*). Four independent verification channels — analytical closed form (D1), Richards–Wolf SIL quadrature (D2), PSD-shaped Monte Carlo (D3), and a decoupled-limit negative control (D4) — agree with the analytical prediction within the tolerance proper to each: D1 exact by construction, D2 within +2.55% of Cheng 2023, D3 with cumulant-truncation residual at the numerical precision floor, and D4 at the decoupled floor $\kappa_{\text{physical}} = 1.000100$ with $\text{cond}(H_C) = 1.0000$.

What Case B delivers beyond Case A and PC1–PC11. (a) a ternary triangle closure replacing the binary bridge identity $S = F$, with three disambiguated tightening symbols and a specific anchor value $\kappa_{\text{target}} = 8.5$; (b) a literature-consistency table with three category-A same-observable rows confirmed and three category-B plausibility rows consistent with the published landscape (Table 3); (c) the H_C identification theorem (Slot ii, Eq. (13)), which proves that $\text{cond}(H_C)_{c=0} = \kappa_{\text{target}}^2$ identically in mixed-budget parameterisation. At the Case B anchor the computed value $\text{cond}(H_C) = 73.01$ decomposes as $\kappa_{\text{target}}^2 + 1.06\%$ cross-coupling surplus (Eq. (14)), and this surplus is the Gap 2 measurement target; (d) Eq. (16) as the quantitative falsification threshold, with ratio band $[0.95, 1.10]$ reflecting the fact that the denominator of that ratio is an identity rather than an independent measurement.

Scope honesty. Case B tests the Gaussian-thermal, strain-free, ensemble-NV, room-temperature, scalar-SIL instance of the triangle. Heavy-tailed thermal PSDs, strain and pressure couplings, single-NV inhomogeneity, cryogenic dD/dT non-linearity, and vector SIL instances are explicitly out of scope and carry forward-pointers. The ensemble-test caveat (Slot v) is disclosed: the inverse-FFT sampler used here is variance-deterministic, so the D3 residual histogram sits at the numerical floor rather than at a physical finite-sample floor.

Deprecation clause. When Gap 2 is closed, Slot (vii) is archived to a “historical foundations” appendix alongside Case A Slot (vii). Built-in obsolescence is by design.

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End of Case B manuscript (v12, 2026-04-21).